

 ISRO Scientist/Engineer 'SC' (Computer Science) – Mock Question Paper 2025

Total Questions: 80 | Marks: 80 | Duration: 90 minutes
(All questions carry 1 mark each. No negative marking.)

Part A – Discipline Specific (60 Questions)

Data Structures & Algorithms

1. The preorder traversal of a BST is (15, 10, 8, 12, 20, 18, 25). The postorder traversal is:

- (a) 8, 12, 10, 18, 25, 20, 15
- (b) 12, 8, 10, 25, 18, 20, 15
- (c) 8, 10, 12, 15, 18, 20, 25
- (d) 25, 18, 20, 15, 12, 10, 8

2. Time complexity of Quick Sort in the best case is:

- (a) $O(n^2)$
- (b) $O(n \log n)$
- (c) $O(n)$
- (d) $O(\log n)$

3. Maximum number of nodes in a complete binary tree of height h is:

- (a) $2^{(h+1)} - 1$
- (b) $2^h - 1$
- (c) h^2
- (d) $h \log h$

4. Minimum number of edges in a connected graph with n vertices is:

- (a) $n - 1$
- (b) n
- (c) n^2
- (d) $n \log n$

5. A hash table uses chaining. In the worst case, time complexity of search is:

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n \log n)$

Computer Networks

6. Which protocol is used in the transport layer?

- (a) IP
- (b) TCP
- (c) ICMP
- (d) ARP

7. A 10 Mbps Ethernet has propagation delay 25 μ s and frame size of 1500 bytes. Minimum transmission time is:

- (a) 1.2 ms
- (b) 1.5 ms
- (c) 1.8 ms
- (d) 2 ms

8. In CSMA/CD, which condition must be met to detect collision?

- (a) Transmission time $> 2 \times$ propagation delay
- (b) Transmission time $<$ propagation delay
- (c) Propagation delay = 0
- (d) None

9. IPv6 address length is:

- (a) 32 bits
- (b) 64 bits
- (c) 128 bits
- (d) 256 bits

10. SMTP is used for:

- (a) File transfer
- (b) Email transfer
- (c) Web browsing
- (d) Routing

Operating Systems

11. Which page replacement algorithm suffers from Belady's Anomaly?

- (a) LRU
- (b) FIFO
- (c) Optimal

(d) Clock

12. Deadlock avoidance can be done using:

- (a) Banker's Algorithm
- (b) FIFO
- (c) Round Robin
- (d) Belady's theorem

13. In FCFS scheduling, if jobs arrive in increasing burst time order, average waiting time will be:

- (a) Minimum
- (b) Maximum
- (c) Zero
- (d) Random

14. Thrashing occurs when:

- (a) CPU utilization is high
- (b) Page fault rate is very high
- (c) Degree of multiprogramming decreases
- (d) None

15. Which is NOT a type of scheduling?

- (a) Long-term
- (b) Medium-term
- (c) Short-term
- (d) Extreme-term

DBMS & SQL

16. In SQL, `SELECT * FROM A, B` is equivalent to:

- (a) Natural join
- (b) Cross join
- (c) Outer join
- (d) Inner join

17. Fifth Normal Form (5NF) deals with:

- (a) Join dependency
- (b) Functional dependency
- (c) Multivalued dependency
- (d) Domain-key dependency

18. Which index in DBMS requires data to be sorted?

- (a) Hash index
- (b) B+ tree index
- (c) Bitmap index
- (d) None

19. In ACID properties, durability ensures:

- (a) Changes persist even after crash
- (b) Transactions are isolated
- (c) Transactions rollback
- (d) None

20. Which SQL query deletes all teachers in departments located in CSE building?

- (a) DELETE FROM teacher WHERE dept_name IN 'CSE';
- (b) DELETE FROM department WHERE building='CSE';
- (c) DELETE FROM teacher WHERE dept_name IN (SELECT dept_name FROM department WHERE building='CSE');
- (d) None

Computer Organization & Architecture

21. A full adder requires:

- (a) 2 inputs, 2 outputs
- (b) 3 inputs, 2 outputs
- (c) 2 inputs, 3 outputs
- (d) 3 inputs, 1 output

22. Instruction pipeline hazards can be due to:

- (a) Structural conflicts
- (b) Data dependency
- (c) Control dependency
- (d) All of the above

23. Cache mapping techniques include:

- (a) Direct mapping
- (b) Fully associative
- (c) Set associative
- (d) All of the above

24. The instruction MOV A, B belongs to which addressing mode?

- (a) Direct
- (b) Immediate
- (c) Register
- (d) Indexed

25. Which of the following is NOT RISC characteristic?

- (a) Fixed instruction size
- (b) Microprogramming
- (c) Load-store architecture
- (d) Large number of registers

Theory of Computation & Compiler Design

26. LL(1) parser uses:

- (a) Rightmost derivation with lookahead 1
- (b) Leftmost derivation with lookahead 1
- (c) Leftmost derivation with no backtracking
- (d) Rightmost derivation with no backtracking

27. In regular expressions, the language $(a|b)^*ab$ represents:

- (a) All strings ending with ab
- (b) All strings starting with ab
- (c) All strings containing ab once
- (d) All strings of length 2

28. The language generated by grammar $S \rightarrow aSb \mid ab$ is:

- (a) $a^n b^n, n \geq 1$
- (b) $a^n b^{2n}, n \geq 1$
- (c) $a^n b^{n+1}, n \geq 0$
- (d) None

29. Which of the following is NOT recursively enumerable?

- (a) Union of two recursive languages
- (b) Complement of recursive language
- (c) Intersection of recursively enumerable and recursive
- (d) Complement of recursively enumerable language

30. In compiler design, DAG is used for:

- (a) Register allocation

- (b) Code optimization
- (c) Syntax analysis
- (d) None

Software Engineering

31. Spiral model is most appropriate when:

- (a) Requirements well-defined
- (b) Risk analysis is important
- (c) Small project
- (d) None

32. Black-box testing focuses on:

- (a) Code logic
- (b) Functionality
- (c) Data structure
- (d) Algorithms

33. Which is NOT a software quality attribute?

- (a) Reliability
- (b) Maintainability
- (c) Debugging
- (d) Portability

34. Function point analysis is used for:

- (a) Software testing
- (b) Software effort estimation
- (c) Scheduling
- (d) None

35. Which UML diagram shows system's static structure?

- (a) Class diagram
- (b) Sequence diagram
- (c) Use-case diagram
- (d) Activity diagram

Programming (C/C++)

36. The size of int in C (on 32-bit machine) is:

- (a) 1 byte
- (b) 2 bytes
- (c) 4 bytes
- (d) 8 bytes

37. What will be output of `printf("%d", 5/2);` in C?

- (a) 2.5
- (b) 2
- (c) 3
- (d) Error

38. Dangling pointer occurs when:

- (a) Memory is freed but pointer is not set NULL
- (b) Pointer is initialized
- (c) Pointer points to valid memory
- (d) None

39. In C++, which operator cannot be overloaded?

- (a) +
- (b) =
- (c) ::
- (d) []

40. Constructor in C++ is invoked when:

- (a) Object is created
- (b) Program ends
- (c) Function ends
- (d) None

Discrete Mathematics / Graph Theory

41. Handshaking lemma states:

- (a) Sum of degrees = 2 × edges
- (b) Degree = edges – 1
- (c) Degree = vertices – 1
- (d) None

42. Girth of a graph is:

- (a) Number of vertices

- (b) Length of shortest cycle
- (c) Degree of graph
- (d) None

43. Euler's formula for planar graph is:

- (a) $V - E + F = 2$
- (b) $V + E - F = 2$
- (c) $V \times E = F$
- (d) None

44. In a bag of 6 blue, 5 white, 4 yellow balls, probability of drawing NOT white ball is:

- (a) $2/3$
- (b) $1/3$
- (c) $1/5$
- (d) None

45. Number of sub-strings in a string of length m is:

- (a) m^2
- (b) $m(m+1)/2$
- (c) $m \log m$
- (d) None

Numerical Methods & Probability

46. Newton-Raphson method is used for:

- (a) Differentiation
- (b) Root finding
- (c) Integration
- (d) None

47. If X is a fair die random variable, $P(X=2 \text{ or } X=4)$ is:

- (a) $1/6$
- (b) $2/6$
- (c) $1/3$
- (d) $1/2$

48. Mean of numbers 5, 10, 15, 20 is:

- (a) 10
- (b) 12.5
- (c) 15

(d) 20

49. Standard deviation is always:

- (a) Positive
- (b) Negative
- (c) Zero or positive
- (d) None

50. For matrix multiplication $A(m \times n) \times B(n \times p)$, complexity is:

- (a) $O(m \times p)$
- (b) $O(m \times n \times p)$
- (c) $O(n^2)$
- (d) $O(p^2)$

Digital Logic

51. A full adder has:

- (a) 2 inputs, 2 outputs
- (b) 3 inputs, 2 outputs
- (c) 2 inputs, 3 outputs
- (d) None

52. NAND gate is:

- (a) Universal gate
- (b) Exclusive gate
- (c) Arithmetic gate
- (d) None

53. A flip-flop is:

- (a) Combinational circuit
- (b) Sequential circuit
- (c) Adder circuit
- (d) None

54. A multiplexer selects:

- (a) One input from many inputs
- (b) Many inputs from one input
- (c) Always two inputs
- (d) None

55. A truth table with n variables has:

- (a) $2n$ rows
- (b) n^2 rows
- (c) n rows
- (d) None

56. A full subtractor requires:

- (a) 2 inputs, 2 outputs
- (b) 3 inputs, 2 outputs
- (c) 2 inputs, 3 outputs
- (d) None

57. A logic circuit that gives LOW when inputs are equal is:

- (a) XOR
- (b) XNOR
- (c) NAND
- (d) NOR

58. Karnaugh map is used for:

- (a) Minimization of Boolean expressions
- (b) Graph drawing
- (c) Matrix simplification
- (d) None

59. Number of states in DFA equivalent to NFA is:

- (a) At most 2^n
- (b) At least 2^n
- (c) n
- (d) None

60. Hamming code is used for:

- (a) Error detection and correction
- (b) Encryption
- (c) Compression
- (d) None

Part B – Mathematics & Aptitude (20 Questions)

61. Probability of getting head in a fair coin toss is:

- (a) 0
- (b) 1
- (c) $1/2$
- (d) None

62. Eigenvalues of matrix $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ are:

- (a) 2,2
- (b) 2,3
- (c) 3,3
- (d) None

63. If sum of first 50 natural numbers is S, then S =

- (a) 1275
- (b) 1225
- (c) 1250
- (d) 1280

64. A man covers $2/5$ distance at 20 km/hr and rest at 30 km/hr. Average speed =

- (a) 24 km/hr
- (b) 25 km/hr
- (c) 26 km/hr
- (d) 28 km/hr

65. Missing number in sequence: 2, 6, 12, 20, ?, 42

- (a) 28
- (b) 30
- (c) 32
- (d) 36

66. If RAM is to MEMORY, then CPU is to:

- (a) Computer
- (b) Processing
- (c) Control
- (d) Execution

67. If 3 men can complete a work in 15 days, 5 men will complete it in:

- (a) 6 days
- (b) 9 days
- (c) 12 days
- (d) 15 days

68. If $\log_{10} 1000 = x$, then $x =$

- (a) 1
- (b) 2
- (c) 3
- (d) 10

69. If $\sqrt{x+9} = 5$, then $x =$

- (a) 16
- (b) 25
- (c) 36
- (d) None

70. A shopkeeper buys a product at ₹100 and sells at ₹120. Profit % =

- (a) 10%
- (b) 15%
- (c) 20%
- (d) 25%

71. Average of 6, 8, 10, 12, 14 =

- (a) 8
- (b) 10
- (c) 12
- (d) 14

72. The next prime after 47 is:

- (a) 49
- (b) 51
- (c) 53
- (d) 59

73. If a train travels 60 km in 1 hour, its speed in m/s =

- (a) 15
- (b) 16.67
- (c) 18
- (d) 20

74. The value of $(81)^{(1/4)} =$

- (a) 2
- (b) 3
- (c) 4

(d) 9

75. Simplify: $(\frac{2}{3} + \frac{3}{4}) =$

(a) $\frac{17}{12}$

(b) $\frac{23}{12}$

(c) $\frac{25}{12}$

(d) $\frac{13}{12}$

76. If a person walks 3 km east, then 4 km north, distance from start =

(a) 5 km

(b) 6 km

(c) 7 km

(d) None

77. If simple interest on ₹1000 for 2 years at 5% = X, then X =

(a) 50

(b) 100

(c) 200

(d) 500

78. If $a:b = 2:3$, $b:c = 4:5$, then $a:b:c =$

(a) 2:3:5

(b) 8:12:15

(c) 10:15:20

(d) None

79. A cube has side 2 cm. Its volume =

(a) 4 cm^3

(b) 6 cm^3

(c) 8 cm^3

(d) 12 cm^3

80. The angle in a semicircle is always:

(a) 30°

(b) 60°

(c) 90°

(d) 120°